

Practical No. 7

Aim & Study the network IP address configuration

Theory : The IP address stands for Internet Protocol address is also called IP number or internet address. It helps us to specify the technical format of addressing a packets scheme. An IP address is a numerical label assigned to the devices connected to a computer network that uses the IP for communication. IP address act as an identifier for a specific machine on a particular network. It also helps us to develop a virtual connection between a source and a destination.

Types of IP address : Public, Private, Static, Dynamic

Public IP Addresses : Public address is an address where one primary address is associated with the whole network. In this type of IP address, each of connected device has the IP address.

Static IP addresses

A static IP address is an IP address that cannot be changed. These are fixed and are manually assigned to a system device. On the network configuration page, the network administrator manually inputs the IP address for every system. Moreover, the static address is not changed.

VLSM	It supports VLSM. Here VLSM means that IPv4 converts IP address into a subnet of diff. sizes.	It does not support VLSM.
Address configuration	It supports manual & DHCP configuration	It supports manual, DHCP auto config & renumbering
Address space	It generates 4 billion unique addresses	It generates 340 undecillion unique addresses
End to end connection integrity	IPv4, end-to-end connection integrity is unachievable	In case of IPv6 end to end connection is achievable
Security features	In IPv4, security depends on appl ⁿ . This IP addresses is not developed in keeping security feature in mind.	In IPv6 IPSEC is developed for security purposes.
Fragmentation	Fragmentation is done by sender & forwarding routers	Fragmentation is done by sender only
checksum	The checksum field is available in IPv4	The checksum field is not available in IPv6.

Practical No. 8

Aim: study the network IP address configuration

Theory: The Internet Protocol version 4 (IPv4) is a protocol for use on packet-switched link layer network. IPv4 provides an addressing capabilities of approximately 4.3 billion addresses. The Internet Protocol version 6 (IPv6) is more advanced & has better features compared to IPv4.

Features	IPv4	IPv6
Address length	IPv4 is a 32-bit address	IPv6 is a 128-bit address
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by dot (C-).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by colon
Classes	IPv4 has 5 different classes of IP address that includes Class A, Class B, Class C, Class D & Class E	IPv6 has a large number of IP addresses.
Number of IP addresses	IPv4 has a limited no. of IP addresses	IPv6 has large no. of IP addresses.

Static IP addressing vs Dynamic IP addressing

Definition	It is a permanent numeric address that is manually issued to a network device	It is a temporary IP address assigned when it connects to network
Provider	It is provided by Internet service Provider (ISP)	It is provided by DHCP (Dynamic Host Config Protocol)
Changes	It doesn't change with time	It may be changed at any time
Device tracking	Devices may be traced easily	Devices may be diff. to trace
Cost	It is expensive to utilize & maintain	It is less expensive to use
Security	It is less secure than the dynamic IP address	It offers high security
Stability	It is highly stable	It is less stable

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23/4/20